

week7

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What I did

- Deeper model tuning
- Investigate feature importance
- 75% of Method section

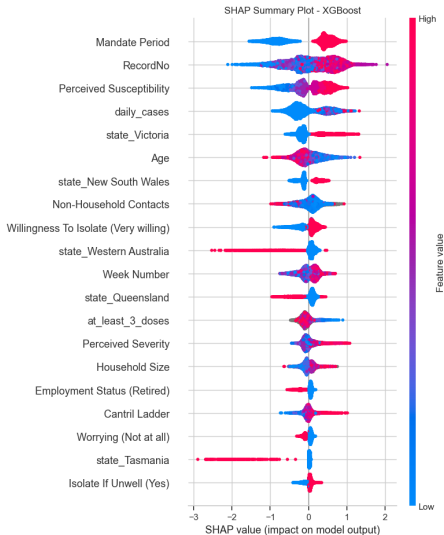
Deeper model tuning

```
grid_mlp = {  
    'lr': [0.0005, 0.001, 0.005],  
  
    # Layer 1 (Always exists)  
    'module__hidden_size_1': [32, 50, 64],  
  
    # Layer 2 (0 means 1-layer network)  
    'module__hidden_size_2': [0, 16, 25, 50],  
  
    # Layer 3 (0 means max 2-layer network)  
    'module__hidden_size_3': [0, 8, 16],  
  
    'module__activation': [nn.ReLU, nn.LeakyReLU, nn.GELU, nn.Tanh],  
  
    'module__dropout_rate': [0.3, 0.4, 0.5],  
  
    'optimizer__weight_decay': [0.0, 1e-4]  
}
```

- The AUCROC is still lower than XGBoost.

Feature impact distribution (SHAP)

- I forgot to remove the record number.



Plan for next week

- Refit the models
- Result section of report

- Should I include detailed theoretical explanations for every model and method applied in this report?

Logistic Regression

Logistic regression is a generalised linear model widely applied in behavioural analysis, that outputs the probability of a target class. Serving as the baseline model, an L2 regularization penalty was introduced to prevent overfitting in the high-dimensional feature space. Specifically, the inverse of regularization strength (parameter C) was searched across a grid of [0.1, 1.0, 10.0].